In [1]:
```
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("ratings.tsv", sep="\t")
print(df.shape)
df.head()
```

```
(15, 5)
```

Out[1]:

	restaurant	food	ambience	service	rating
0	1	85	82	89	78
1	2	80	90	80	85
2	3	83	86	83	85
3	4	70	96	75	72
4	5	68	80	78	75

In [2]:
```
features = ["ambience", "food", "service"]
X = df[features]
y = df["rating"]

model = LinearRegression()
model.fit(X, y)

print("Intercept:", round(model.intercept_, 4))
print("R^2:", round(model.score(X, y), 4))
print("coef_ (in column order):", model.coef_.round(4))
```

```
Intercept: -15.8713
R^2: 0.7943
coef_ (in column order): [0.1338 0.7338 0.324 ]
```

In [3]:
```
ranked = (
    pd.DataFrame({"feature": features, "coefficient": model.coef_})
    .sort_values("coefficient", ascending=False)
    .reset_index(drop=True)
)
ranked.index += 1
ranked
```

Out[3]:

	feature	coefficient
1	food	0.733776
2	service	0.323981
3	ambience	0.133761

Standardized-coefficient cell to check that the ranking holds

In [4]:
```
std_coefs = pd.Series(
    model.coef_ * X.std().values / y.std(), index=features
).sort_values(ascending=False)
std_coefs.round(4)
```

Out[4]:
```
food        0.5783
service     0.3143
ambience    0.1574
dtype: float64
```

recommendation

Food quality is most heavily associated with score, so restaurants looking to raise their score should focus on food quality first. Every point of food score is associated with roughly 0.73 points in terms of rating holding ambience and service constant, which is more than twice the effect of service (0.32) and over five times that of ambience (0.13). This is based on only 15 restaurants and on association, so the ranking should be treated as a strong suggestion not a guarantee.